

Advanced Architectures and Strategic Remediation of Gaps in Modern Sentiment Analysis

The landscape of natural language processing has undergone a fundamental transformation between 2024 and 2026, transitioning from basic polarity detection to a sophisticated discipline of intent engineering. Modern sentiment analysis is no longer confined to the binary classification of "positive" or "negative" text; instead, it has evolved into a multi-dimensional field that incorporates emotional nuance, cultural context, and domain-specific linguistic shifts.[1, 2, 3] Despite the rapid advancement of large language models, significant gaps remain that frequently lead to the failure of production-grade systems.[1, 4] These failures are often rooted in the inability of models to handle sarcasm, the drift in linguistic patterns such as social media slang, and the inherent fragility of general-purpose models when applied to high-stakes domains like finance and healthcare.[5, 6, 7] Addressing these gaps requires a move away from massive, generalized API calls toward modular, specialized architectures that prioritize data integrity and context-aware reasoning.[1, 8, 9]

The Sarcasm and Irony Gap: From Lexicons to Multi-Head Architectures

The traditional binary of sentiment analysis—positive versus negative—has encountered a formidable obstacle in the pervasive nature of digital irony. In 2026, sarcasm is recognized as a primary dialect of the internet, necessitating a shift from simple keyword lexicons to complex intent engineering.[1] The mechanism of sarcasm involves a systematic inversion of polarity, where highly positive lexical choices are employed to convey profound dissatisfaction.[5] For instance, the phrase "Truly a masterpiece of planned obsolescence" contains positive descriptors like "masterpiece" but carries a biting negative intent.[1] To address this gap, researchers have proposed Dual-Head Architectures. In these systems, a transformer-based encoder produces hidden states that are passed to two distinct classification heads: one optimized for sentiment polarity and another for sarcasm detection.[1, 10]

The technical implementation of these architectures often involves a shared transformer backbone, such as RoBERTa or Llama-2, which is fine-tuned to capture the contextual relationships that signal incongruity.[10, 11] By training these heads simultaneously, the model learns that the presence of high-intensity positive words combined with a high sarcasm probability should result in an "Extremely Negative" final output.[1] This multitasking approach allows the model to develop a more nuanced understanding of incongruity—the delta between literal and intended meaning.[10, 12] Research utilizing the Self-Annotated Reddit Corpus (SARC) has demonstrated that multi-head attention mechanisms are superior at capturing the subtle, context-dependent cues of sarcasm compared to traditional sequential models.[10, 11]

Sarcasm Detection Element	Description and Role	Technical Mechanism
Hyperbolic Features	Detects intensifiers and interjections (e.g., "totally," "wow") that signal exaggeration.	Feature extraction via Word2Vec or hyperbolic embeddings.[12]

Incongruity Learning	Identifies semantic mismatches between literal text and underlying sentiment.	Multi-level Incongruity Fusion (AMIF) and co-attention layers.[13]
Multi-Modal Fusion	Cross-references text sentiment with visual cues in images or videos.	Early fusion of text and image embeddings in a unified transformer.[1, 14]
Contextual Memory	Analyzes previous turns in a conversation to determine the speaker's true intent.	Long Contextual Memory in frontier LLMs or graph-based memory (Mem0).[2, 15]

The challenge is further magnified in multimodal settings, where sarcasm is triggered by mismatches between text and images.[13, 14] For example, a "Happy" caption paired with a "Frustrated" facial expression requires a fusion strategy that can interpret inherent ambiguity within both modalities.[13] The development of MMSD (Multimodal Sarcasm Detection) benchmarks and frameworks like CIRM (Multi-image Sarcasm Reasoning) demonstrates a shift toward modeling these complex affective relations across multiple images and text spans.[13, 14] In 2025, tools like Brandwatch began incorporating comprehensive NLP engines capable of detecting dozens of distinct emotional states, moving beyond the traditional three-class classification toward a more granular emotional spectrum.[3]

Linguistic Drift and the Tokenization Barrier

Language evolves at a pace that frequently outstrips the update cycles of static language models.[1, 16] Between 2024 and 2026, a surge in new slang terms—such as "cooked," "delulu," and "slop" (referring to AI-generated garbage)—has created a significant performance gap for models using older tokenizers.[1] When a model encounters a word it does not recognize, it defaults to an [UNK] (Unknown) token, effectively losing all emotional weight and context associated with that term.[1] This "Tokenization Drift" is a primary reason why basic sentiment analysis models, even those based on large language models, can fail to grasp the intensity of modern consumer feedback.[1]

The remediation for this gap involves Custom Tokenizer Extensions.[1] Instead of relying on a pre-trained tokenizer from 2022 or 2023, developers must now implement workflows that add current slang and domain-specific jargon to the model's vocabulary before fine-tuning.[1] This ensures that the model treats these terms as distinct tokens with their own learned embeddings rather than discarding them as noise.[1] Furthermore, the preservation of emojis has become a critical preprocessing requirement.[5] Emojis are not merely decorative; they often act as the final arbiter of sentiment.[1] A sentence like "I love sitting in traffic" might be classified as positive by a simple model, but the addition of a 🦴 (skull) or 🤡 (clown) emoji completely flips the polarity.[1] In addition to slang, the rise of code-switching—where users alternate between

multiple languages within a single post—poses a structural challenge for sentiment models.[17, 18, 19] Monolingual models often struggle with the grammatical complexities and unpredictable switch points found in code-mixed data from multilingual communities.[17, 20] The current state-of-the-art approach to this problem involves the use of multilingual transformers like mBERT (Multilingual BERT) or specialized pipelines like BiLingua, which are designed to capture syntactic structure across mixed-language inputs.[17, 19, 20] These models leverage cross-script knowledge transfer to understand text written in different scripts, such as Roman Urdu mixed with English.[20, 21]

Tokenization Challenge	Impact on Sentiment Accuracy	Strategic Fix
Unknown Tokens ([UNK])	Loss of sentiment signal for new slang or jargon.	Vocabulary expansion and custom tokenizer fine-tuning.[1]
Emoji Stripping	Misses critical emotional intensity and sarcasm cues.	Conversion of emojis to specific sentiment tokens or preservation of Unicode.[5]
Code-Mixing Noise	Confusion between language-specific grammars in mixed text.	Multilingual transformers (mBERT) and language identification gates. [19, 20]
Contraction Ambiguity	Fails to detect negation (e.g., "don't" vs "do").	Rule-based expansion of contractions during preprocessing.[5]

Modern preprocessing sequences now prioritize the removal of metadata, lowercasing, and the expansion of acronyms before the tokenization stage.[22, 23] However, BERT-based models typically perform best with minimal preprocessing, as their architecture is designed to process raw text with rich linguistic variation.[5, 23] The integration of dependency parsing further enhances these models by identifying the "negation scope"—how many words following a negation cue are affected—thereby preventing the misclassification of phrases like "not very good" as positive.[5]

Domain Sensitivity and the Fragility of Generalization

One of the most significant gaps in current sentiment analysis is the lack of domain transferability.[5] A model trained on movie reviews or social media data will frequently fail when applied to specialized fields like finance, healthcare, or legal documentation.[5, 24, 25] This failure occurs because sentiment cues are highly context-dependent.[5, 26] For example, the term "unpredictable" signals a positive sentiment in a review for a thriller movie but represents a highly negative sentiment when used to describe the performance of car brakes or financial asset volatility.[5]

In the financial sector, sentiment analysis is increasingly used for algorithmic trading and risk assessment, where even a minor shift in polarity can trigger

significant market movements.[7, 27] General-purpose models often struggle with "financial-speak" and the misalignment between informal user-generated content (e.g., financial tweets) and structured reference data (e.g., analyst reports).[6, 7] To address this, industry-specific models like FinBERT have been developed to quantify sentiment within the precise linguistic context of financial markets.[7] These models are often integrated into hybrid frameworks, such as the combination of Long Short-Term Memory (LSTM) networks and Random Forests, to capture non-linear dependencies and the "Granger cause" relationship between social media sentiment and short-term stock price fluctuations.[7]

Industry Sector	Primary Sentiment Challenge	Recommended Architecture
Financial Markets	High sensitivity to jargon and market-moving events.[7]	FinBERT integrated with sequential learners (LSTM).[7]
Healthcare	Complex medical narratives and implicit patient emotions.[25]	Multi-scale BERT and dimensional (Valence-Arousal) models.[6, 25]
Public Health	Detection of vaccine hesitancy and complacency.[28]	LLMs fine-tuned on public health communication datasets.[28]
Legal Services	High-stakes decision-making and extreme bias risks.[6, 29]	Domain-adapted SciBERT or specialized LLMs for law.[6]

Healthcare sentiment analysis faces its own unique hurdles, including the complexity of discourse in medical narratives and the presence of low-arousal writing styles.[24, 25] Unlike the high-energy language often found on social media, doctors' self-reflection texts or patient medical narratives may express emotions implicitly.[25] The shift from categorical sentiment (positive/negative) to dimensional sentiment—measuring valence (pleasantness) and arousal (activation)—has been proposed as a more accurate way to capture these subtle affective states in medical contexts.[25] Furthermore, the lack of annotated datasets for low-resource languages in healthcare remains a barrier to global deployment.[20, 25]

The Granularity Gap: Moving Toward Aspect-Based Sentiment Analysis (ABSA)

General sentiment scores are increasingly viewed as insufficient for actionable business intelligence.[1] A "Neutral" average score for a review like "The UI is gorgeous, but the load times are a nightmare" is misleading because it obscures two very different and intense sentiments.[1] Aspect-Based Sentiment Analysis (ABSA) resolves this by decomposing a single document into specific attributes, such as "Design/UI" and "Performance," and assigning a sentiment polarity to each.[1, 30, 31]

The technical implementation of ABSA in 2026 involves the use of dependency parsing to link adjectives directly to specific nouns.[1, 32] This ensures that the sentiment of "gorgeous" is correctly associated with "UI" and "nightmare" is associated with "load times".[1] Modern ABSA models often combine dependency trees with self-attention mechanisms to generate weight matrices that emphasize the locational information of aspect terms.[32] This is particularly critical for long-form content, where multiple aspects may be discussed across several paragraphs.[31, 32]

ABSA Task	Objective	Methodology in 2025/2026
Aspect Extraction	Identifying explicit and implicit entities (e.g., "battery," "speed").	Graph Convolutional Networks (GCN) and dependency trees.[30, 32]
Sentiment Classification	Assigning polarity to each extracted aspect.	Fine-tuned transformers or hybrid rule-based pipelines. [30, 31]
Opinion Term Extraction	Finding the specific words expressing the sentiment.	Multi-task learning and self-attention weight matrices.[31, 33]
Aspect-Sentiment Alignment	Mapping the correct sentiment to the correct aspect span.	Dependency parsing and local-focused GCNs.[30, 32]

Implicit aspect extraction—where the aspect is not explicitly named but can be inferred from context—remains a frontier in ABSA research.[33] For example, a sports review saying "he couldn't keep up in the final quarter" implies a negative sentiment toward the aspect of "stamina" or "fitness".[33] Large language models show promise in this area but are prone to inconsistencies and require expert-revised annotation sets to achieve high precision.[33, 34] For long-form documents, the computational cost of performing ABSA across every sentence can be prohibitive, leading to the adoption of hybrid pipelines that use rules for aspect detection and neural models for classification.[30]

Efficiency, Scale, and the Rise of Small Language Models (SLMs)

While large language models (LLMs) like GPT-4 and Claude 3.5 offer exceptional reasoning capabilities, their deployment for high-volume sentiment analysis is often hindered by high latency and prohibitive costs.[1, 8, 9] For many organizations, processing millions of social media posts or customer support tickets using frontier LLMs is economically unsustainable, with costs ranging from \$8,000 to \$12,000 per million posts.[9] Furthermore, the latency of cloud-hosted LLMs, which can be hundreds of milliseconds per token, makes them unsuitable for real-time applications where sub-second response times are required.[8, 9]

In response to these gaps, the industry has pivoted toward Small Language

Models (SLMs) and Parameter-Efficient Fine-Tuning (PEFT).[1, 8, 35] Models such as DistilRoBERTa-v2, Phi-3, and Mistral 7B offer a more favorable balance between accuracy and efficiency.[1, 8] These models can often be fine-tuned on standard consumer GPUs in under 15 minutes using techniques like LoRA (Low-Rank Adaptation), which modifies only a tiny fraction of the model's parameters.[1]

Model Scale Type	Parameter Range	Token Latency	Relative Inference Cost	Optimal Application
Frontier LLM	100B - 1T+	500 - 800 ms	100x Base	Complex reasoning, research.[8]
Mid-Sized LLM	15B - 70B	200 - 400 ms	10x Base	High-accuracy general analysis.[8, 36]
Small LM (SLM)	1B - 15B	50 - 150 ms	1x Base	Real-time edge deployment. [8, 35]
Specialized Classifier	< 1B	10 - 30 ms	0.1x Base	High-volume binary filtering.[1, 9]

The optimal architecture for a 2026 sentiment project is a Two-Stage Cascade. [8, 9] In the first stage, a lightweight "Relevance Gate" (e.g., DistilBERT) filters out noise and irrelevant content at extreme speeds.[9] Only the remaining high-value or ambiguous records are then routed to a larger LLM for deep contextual analysis.[9] This approach can reduce total processing costs by 80% while maintaining an overall accuracy of over 90%.[9] Furthermore, SLMs run locally, which enhances data privacy for sensitive applications in healthcare and finance.[8, 35]

The Synthetic Data Trap: Preventing Model Collapse

As AI-generated content increasingly floods the internet, sentiment analysis models face the risk of "Model Collapse".[1, 37, 38] This phenomenon occurs when a model is trained recursively on its own output or the output of other models, leading to a loss of the original human data distribution.[38, 39] Over time, these models lose their ability to detect the "long tail" of raw human emotion, resulting in outputs that are bland, paraphrased, and increasingly disconnected from reality.[37, 38] In extreme cases, this leads to "Model Autophagy Disorder" (MAD), where the model's performance on real-world data degrades significantly even if it continues to perform well on synthetic benchmarks.[38, 39]

The remediation for model collapse is a rigorous Human-in-the-Loop (HITL)

data strategy.[1, 37, 40] In 2026, the industry has realized that high-quality, human-annotated data is a finite and precious resource.[38] Organizations are now advised to cap synthetic data ratios in their training sets and prioritize the licensing of verified, human-written text for sensitive domains like health and finance.[38] Furthermore, data source provenance must be tracked ruthlessly, labeling every record as human-authored, human-edited, or synthetic.[38]

Data Provenance Category	Reliability Level	Impact on Model Diversity	Recommended Strategy
Human-Authored	Highest	Preserves the long-tail and niche emotions.	Use as the primary training foundation.[38]
Human-Edited Synthetic	High	Balances scale with factual correctness.	Ideal for task-specific fine-tuning.[38]
Purely Synthetic	Medium-Low	Risks recursive "sameness" and blandness.	Cap at 10-20% of total training data.[38]
Unverified Web Crawl	Lowest	High risk of pollution from AI "slop".	Harden crawl policies and use whitelists.[38]

Data quality remains the dominant barrier to AI success, with 64% of organizations citing it as their top challenge.[4] Poor data integrity not only undermines model accuracy but also has massive economic impacts, with historical estimates suggesting trillions of dollars in annual costs due to flawed decision-making and operational inefficiencies.[4] In sentiment analysis, this quality gap manifests as "AI solutions that are almost right, but not quite," leading to time-consuming debugging and a lack of trust in automated insights. [41]

Mitigation of Hallucinations and Grounding Techniques

Large language models are prone to "Emotional Hallucinations," where they invent sentiments or cite non-existent reasons for a user's emotional state.[42, 43, 44] These hallucinations can arise from data biases, ambiguity in prompts, or the probabilistic nature of the model's next-token prediction.[44, 45] In high-stakes environments like scientific peer review or medical diagnosis support, these errors can undermine the credibility of the entire system.[42, 43, 46]

Retrieval-Augmented Generation (RAG) has emerged as a critical technique for mitigating these hallucinations.[43, 44, 47] By anchoring the model's response in verifiable external documents, RAG ensures that sentiment evaluations are grounded in specific facts rather than internal model biases.[44, 47] For example, in the peer review process, a RAG-enabled system can identify the limitations of a paper by explicitly referring to the existing body of scientific literature.[43]

Key strategies for RAG-based sentiment grounding include:

1. **Context Injection:** Providing sufficient information frameworks within the prompt to remove ambiguity.[46, 48]
- 2.
3. **Self-Reflection (Reflexion):** Utilizing LLMs acting as actors, evaluators, and self-reflectors to critique and revise outputs.[45, 48]
- 4.
5. **Multi-Agent Verification:** Employing multiple agents that debate and challenge one another's claims to filter out falsehoods.[48]
- 6.
7. **Chain-of-Verification (CoV):** Forcing the model to generate intermediate checks and reasoning steps before the final answer.[45, 48]
- 8.

Mathematically, hallucination detection often relies on log-probability calculations.[44] When an LLM hallucinates, it often produces words or phrases that are statistically abnormal within the given context, resulting in a lower overall sequence log-probability score.[44] Monitoring these scores provides a quantitative metric for identifying potential grounding failures. The confidence score C for a sequence can be represented as:

$$C = \frac{1}{N} \sum_{i=1}^N \log P(x_i | x_{<i})$$

where N is the number of tokens and P is the probability of each token given the preceding context.[44] Low values for C serve as an automated trigger for human review or rejection of the output.[37, 44]

Multilingual and Code-Switching Architectures

The spontaneous and continuous switching between languages, known as code-switching, presents a profound challenge for sentiment analysis in multilingual digital environments.[17, 18] Users in regions like India, Pakistan, and Central Asia frequently mix regional languages with English in Romanized scripts.[19, 20] Traditional models trained primarily on native scripts struggle with this "cross-script" mismatch.[21]

The primary architectural solution for 2025 is the use of multilingual transformers like mBERT (Multilingual BERT) which are fine-tuned on code-mixed datasets specifically created for these language pairs.[19, 20] These models are designed to capture semantic meaning and contextual nuances regardless of the specific language being used at any given point in the sentence.[19]

Language Pair	Code-Switching Context	Technical Approach
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Roman Urdu/Punjabi-English	Social media comments and feedback.[20]	CMSA-mBERT with implicit contextual embeddings.[20]
Tamil-Tulu-English	Regional forums and Twitter.[19]	mBERT vs. traditional machine learning benchmarks.[19]
Kazakh-Russian	Spontaneous colloquial interactions. [18]	Synthetic data generation for low-resource pairs.[18]
Nepali-English	Native vs. second language expression. [21]	Multi-task learning (MTL) and soft parameter sharing.[21]

Research indicates that speakers often prefer their native language when expressing negative sentiments, while using a second language for neutral or positive statements.[21] This linguistic behavior can be leveraged in model training to improve sentiment extraction accuracy.[21] Furthermore, the use of Machine Translation (MT) in the preprocessing pipeline ensures consistency in semantic understanding across non-English inputs, although this "shortcut" often underperforms compared to natively trained multilingual models.[9, 10]

Ethics, Bias, and Intersectional Fairness

Sentiment analysis models frequently inherit and amplify human biases present in their training data.[2, 49, 50] These biases are often intersectional, meaning they affect specific combinations of demographic traits—such as gender and race—more severely than they do either category in isolation.[51] For example, a model may perform moderately well for women or Black individuals separately but fail when these identities intersect, due to compounded biases and the lack of representative data for "Female+Black" subgroups.[51]

The remediation of these gaps requires architectural improvements at every stage of the model lifecycle:

1. **Pre-model Strategies:** Curating and augmenting datasets using T5-based generation to target underrepresented demographic intersections.[51, 52]
2. **Intra-model Techniques:** Incorporating fairness-aware loss functions, such as those based on Kullback–Leibler (KL) divergence, which penalize biased outputs while maintaining classification performance. [52, 53]
3. **Post-model Adjustments:** Calibrating outputs and using "Equalized Odds" to ensure that the probability of a positive outcome is the same across different groups.[52]

The unified loss function for a bias-aware model can be expressed as:

$$L = L_{task} + \lambda L_{bias} + \mu L_{feedback}$$

integrates signals from human-in-the-loop review.[53] This framework ensures that the model dynamically adapts to real-world contexts, enhancing both accuracy and fairness.[53]

Bias Type	Origin in Sentiment Models	Recommended Mitigation
Historical Bias	Past prejudices reflected in training data.	Data filtering and fairness audits.[50]
Sample Bias	Unrepresentative training populations.	Resampling (SMOTE, GANs) and augmentation.[50, 52]
Amplification Bias	AI exacerbating learned human biases.	Fairness-aware loss functions and regularization.[50, 52]
Ontological Bias	Western-centric worldviews in concepts.	Inclusion of non-Western knowledge and diverse annotators.[5, 50]

Addressing intersectionality requires building datasets that capture real-world language diversity and evaluating performance across combined demographic groups.[51] The outcome of ML algorithms can change as they learn or as training data changes, necessitating continuous monitoring of model metrics for specific groups over time.[50]

Future Outlook: Agentic AI and Emotional Journey Mapping

As we move toward 2026, sentiment analysis is integrating more deeply with Agentic AI and customer journey mapping.[16, 54] Businesses are shifting from retrospective analysis—understanding what happened—to predictive emotional intelligence.[16] Advanced systems now track how a customer's sentiment evolves throughout their entire relationship with a brand, identifying the specific "micro-interactions" that cause feelings to shift.[16] These technologies can detect hesitations, subconscious reactions, and even daily mood fluctuations to personalize the entire user experience.[16] This evolution is supported by new architectures like Mem0, which uses graph-based memory to enhance long-term conversational coherence.[15] Instead of treating each interaction as an isolated event, these systems maintain a "Contextual Continuum" that considers historical coverage and related topics to improve prediction accuracy.[2, 55] Furthermore, the integration of voice and video analysis—detecting emotional states through pitch, rhythm, and facial cues—is projected to improve sentiment detection accuracy by up to 25% over text analysis alone.[2, 55]

Trend for 2026	Impact on Sentiment Analysis	Core Technology
Emotional Journey Mapping	Tracks emotional shifts over long-term customer relationships.[16]	Graph-based memory, temporal resolution mapping.[15, 55]
Multimodal Synergy	Combines text, speech, and visual embeddings for deeper interpretation. [2]	AMIF networks, cross-modal attention.[13, 55]
Personalized Sentiment	Tailors models to individual or group-specific linguistic patterns.[2]	Few-shot prompting, user-aware modulation.[26, 46]
Real-time Reputational Alerting	Identifies threats and automatically prioritizes resources. [9, 16]	Two-stage cascades, high-throughput SLMs.[9, 35]

The global sentiment analysis market is projected to reach \$5.2 billion by 2026, driven by this transition toward empathetic and predictive AI systems.[2, 55] Organizations that successfully bridge the gaps in sarcasm detection, domain specificity, and efficiency will be better positioned to understand the "irrationality" of human behavior and predict market movements with unprecedented precision.[7, 16]

Actionable Recommendations for Project Integration

For the current project, the analysis of these gaps suggests a specific roadmap for building a robust sentiment analysis system that avoids the pitfalls of generic, outdated models. The primary recommendation is the abandonment of a monolithic LLM-only approach in favor of a modular, tiered architecture that addresses context, cost, and accuracy simultaneously.

- 1. Implement a Two-Stage Classification Architecture:** Use a specialized, high-speed classifier (e.g., DistilBERT or a refined DistilRoBERTa-v2) as a "Relevance Gate" to filter noise at low cost.[1, 9] Route only the "ambiguous middle" or high-stakes records to a more capable model.[5, 9]
- 2. Solve the Sarcasm Gap with Dual-Head Design:** Integrate a dedicated sarcasm detection layer. This is particularly critical for social media data where irony is a dominant tone.[1, 10] Training these heads jointly prevents the systematic polarity inversion that plagues standard models.[1, 10]
- 3. Prioritize Granularity through ABSA:** Move beyond aggregate scores. Use dependency parsing to associate sentiments with specific

aspects of the product or service.[1, 30, 32] This provides actionable data for specific internal teams rather than a generic sentiment label. [1, 30]

4. **Extend the Tokenizer for Modern Slang:** Do not use out-of-the-box tokenizers from previous years. Implement a custom tokenizer extension that includes 2025/2026 slang and preserves emojis as sentiment-bearing tokens.[1, 5]
5. **Ground LLMs with RAG and Multi-Agent Verification:** To prevent hallucinations and ensure that the model does not invent emotional subtext, use Retrieval-Augmented Generation to provide the model with a verified context.[44, 45, 47]
6. **Maintain Data Integrity via HITL:** Avoid relying purely on synthetic data for training or fine-tuning. Implement a Human-in-the-Loop strategy for at least 10% of the trickiest cases to provide a "ground truth" that protects the model from collapse and model autophagy disorder.[5, 37, 38]

By focusing on these specific technical remediations, the project can bridge the gap between benchmark performance and real-world reliability, delivering a sentiment analysis system that is not only accurate but also economically and ethically sustainable in the evolving landscape of 2026.

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